Kohlberg's Framework for AI Benchmarking

An annotated bibliography and source-quality guide for the Gemini deep-research report

Companion to: *The Cognitive Architecture of Morality: Kohlberg's Framework and its Application in Artificial Intelligence Benchmarking*

45 CITED SOURCES / OCTOBER 2026

READING GUIDE

The source base is broad, but uneven

The report connects developmental psychology, care ethics, language-model evaluation, and alignment engineering. Its strongest evidence comes from peer-reviewed papers, canonical preprints, and first-party policy documents. Many other citations are best treated as orientation, mirrors, or discovery leads.

45
works cited, annotated in report order

4
Grade A sources: peer-reviewed or authoritative first-party material

13
Grade B sources: credible research, preprints, or institutional context

28
Grade C-D sources: explainers, mirrors, commentary, or weak discovery links

How to use the annotations

Each entry states what the source contributes, what it can reasonably support, and where caution is warranted. Grades evaluate fitness for a research-grade account of Kohlbergian AI benchmarking, not the intrinsic quality of the writing.

Grade A	Peer-reviewed research or a first-party document that is authoritative for its own stated policy or method.
Grade B	Credible preprint, research-group resource, technical text, or institutional source; verify version and scope.
Grade C	Useful secondary explanation or commentary; cite the underlying original for substantive claims.
Grade D	Unvetted upload, discussion thread, aggregator, commercial-help page, polemic, or nonspecific index.

> **Key caution:** A model's high score on a human moral-development instrument shows performance under that instrument. It does not establish consciousness, empathy, moral agency, or a human-like developmental trajectory.

EVIDENCE MAP

Prioritize original methods and stable records

The quickest way to strengthen the report is to promote Grade A sources, use Grade B sources with explicit caveats, and replace Grade C-D citations whenever a primary work exists.

NO.	SOURCE	GRADE	TYPE

Start with 19

Peer-reviewed multilingual DIT evidence

Then 24

Three-dimensional benchmark paper and code

Add 29

Structured moral-reasoning evaluation

Use 41 + 44

Collective process plus first-party constitution

Recommended citation cleanup

- Replace mirrors (items 5, 14, 15, and 32) with canonical publisher, DOI, or arXiv records.
- Collapse duplicate LLM Ethics Benchmark entries (24, 32, and 33) into one canonical citation.
- Replace search and listing pages (22 and 45) with the specific papers they were meant to identify.
- Use tertiary explainers (1-13) for orientation only; add Kohlberg, Gilligan, Rest, and modern review literature directly.
- Keep commentary (16, 39, 40, and 43) clearly separate from empirical results.

> **Interpretive discipline:** Benchmark scores are conditional on prompts, translations, model versions, decoding settings, and scoring rules. Preserve those details whenever numerical comparisons are repeated.

WORKS CITED / 01-03

Foundations of moral development and care ethics

01 Lawrence Kohlberg's Stages of Moral Development

Seattle Anxiety Specialists / Professional explainer **GRADE C**

A clinician-facing overview of Kohlberg's levels and stages. It is useful for plain-language orientation and examples, but it is not a primary research source and should not support claims about universality, developmental sequence, or psychometric validity by itself.

https://seattleanxiety.com/lawrence-kohlbergs-stages-of-moral-development

02 Stages of Moral Development - Lawrence Kohlberg

Educational Technology / Educational explainer **GRADE C**

A concise educational summary of the six-stage model and the role of moral dilemmas. Use it to clarify terminology or prepare teaching material; for historical and empirical claims, replace it with Kohlberg's original work or a scholarly review.

https://educationaltechnology.net/stages-of-moral-development-lawrence-kohlberg/

03 Kohlberg's 6 Stages of Moral Development, With Examples

Structural Learning / Educational blog **GRADE C**

An accessible stage-by-stage treatment with applied examples. Its strength is pedagogy rather than evidence, so it is most appropriate as a reader aid, not as authority for the theory's cross-cultural reach or the prevalence of higher-stage reasoning.

https://www.structural-learning.com/post/kohlbergs-moral-development-stages

Foundations of moral development and care ethics

04 Carol Gilligan - In a Different Voice

Copernican Revolution / Secondary commentary **GRADE C**

An interpretive introduction to Gilligan's critique of justice-centered moral development. It helps explain why care, relationship, and context complicate Kohlberg's hierarchy, but the book itself and peer-reviewed commentary remain the better sources for precise claims.

https://copernicanrevolution.org/psychology-of-gender/in-a-different-voice

05 Kohlberg's Stages of Moral Development

ResearchGate / Repository copy / unclear provenance **GRADE D**

A hosted document about Kohlberg's stages whose bibliographic status is not evident from the citation as given. Treat ResearchGate as a discovery or access route only: verify the author, venue, publication date, and version before relying on its content.

https://www.researchgate.net/publication/387751558_Kohlberg's_Stages_of_Moral_Development

06 Lawrence Kohlberg's stages of moral development

Wikipedia / Tertiary reference **GRADE C**

A broad orientation page covering the stage model, dilemmas, and major criticisms. It can help locate named concepts and references, but its open-editing model makes it unsuitable as the final citation for contested historical, empirical, or cross-cultural claims.

https://en.wikipedia.org/wiki/Lawrence_Kohlberg%27s_stages_of_moral_development

Foundations of moral development and care ethics

07 Lawrence Kohlberg's stages of moral development

Encyclopaedia Britannica / Edited encyclopedia **GRADE B**

A concise, editorially reviewed overview of Kohlberg's theory. It is a credible tertiary source for definitions and historical context, though specialist questions about measurement, gender bias, and stage invariance still require primary or review literature.

https://www.britannica.com/science/Lawrence-Kohlbergs-stages-of-moral-development

08 Kohlberg Heinz Dilemma in Nursing

NurseMyGrade / Essay-help site **GRADE D**

An applied explanation of the Heinz dilemma framed for nursing coursework. It may suggest classroom examples, but commercial academic-help material is not a reliable evidentiary base; use an original instrument, textbook, or peer-reviewed nursing-ethics source instead.

https://nursemygrade.com/kohlberg-heinz-dilemma-in-nursing

09 Gilligan's Ethics of Care: Challenging Kohlberg's Bias

Psychology Notes HQ / Educational blog **GRADE C**

A readable comparison of justice-oriented and care-oriented accounts of moral development. It is valuable for conceptual contrast, but its simplified presentation should be checked against Gilligan's book and later scholarship that questions strict gender essentialism.

https://www.psychologynoteshq.com/kohlberg-vs-gilligan/

Foundations of moral development and care ethics

10 In a Different Voice

Wikipedia / Tertiary reference **GRADE C**

An orientation page on Gilligan's book, its reception, and its place in feminist psychology. Use it to identify themes and leads, not as a substitute for the book or scholarly assessments of the empirical status of care-versus-justice differences.

https://en.wikipedia.org/wiki/In_a_Different_Voice

11 Gilligan's Ethics of Care Explained

Scribd / User-uploaded document **GRADE D**

A user-uploaded summary of Gilligan's framework. Because authorship, version, and editorial review are unclear, it should not carry substantive claims; retain only as a lead, and trace any useful assertion to a citable original source.

https://www.scribd.com/document/808519749/Gilligan

12 Gilligan's Ethics of Care: 3 Stages & the Different Voice

Yu-kai Chou / Popular explainer **GRADE C**

A popular account of Gilligan's developmental sequence and care orientation. It is useful for accessible language and examples, while scholarly work is needed to substantiate the sequence, its relationship to Kohlberg, and claims about sex or gender.

https://yukaichou.com/behavioral-analysis/gilligan-care-ethics-different-voice-moral-development/

Foundations of moral development and care ethics

13 The BEST: In a Different Voice

Tradition Online / Review / commentary **GRADE C**

A reflective review of Gilligan's influential book from a particular intellectual community. It can illuminate reception and interpretation, but it is commentary rather than neutral evidence about the theory's validity or current psychological consensus.

https://traditiononline.org/the-best-in-a-different-voice/

Kohlberg applied to AI and multilingual DIT studies

14 Cognitive morality and AI: A proposed classification of AI systems using Kohlberg's theory of cognitive ethics

ResearchGate / Scholarly-paper mirror **GRADE B**

A conceptual proposal mapping levels of artificial cognition to Kohlberg-like moral stages. It directly supports the report's pattern-identification, rule-abstraction, and meta-pattern analogy, but the analogy is theoretical and should not be mistaken for evidence that models undergo human development.

https://www.researchgate.net/publication/366016044_Cognitive_morality_and_AI_A_proposed_classification_of_AI_systems_using_Kohlberg's_theory_of_cognitive_ethics

15 Constitutional AI: An Expanded Overview of Anthropic's Alignment Approach

ResearchGate / Scholarly-paper mirror **GRADE C**

A secondary overview of Constitutional AI hosted on ResearchGate. It may organize the method's components, but the report should privilege Anthropic's original Constitutional AI paper and first-party technical documentation for training details and evaluation results.

https://www.researchgate.net/publication/391400510_Constitutional_AI_An_Expanded_Overview_of_Anthropic's_Alignment_Approach

16 The moral paradox of LLM technology

AI Security & Governance / Commentary **GRADE C**

An argument about tensions between apparently moral model behavior and the mechanisms that produce it. The piece is useful for framing questions about simulation, agency, and accountability, but its claims are interpretive and need empirical support from model evaluations.

https://gaks.ai/blog/moral-paradox-llm-technology

Kohlberg applied to AI and multilingual DIT studies

17 The Possibility of Discovering Moral Meta-Patterns in AI Systems

arXiv / Preprint **GRADE B**

A speculative research paper on whether sufficiently capable systems might identify higher-order regularities across moral traditions. It is relevant to the report's ethical-resonance hypothesis, yet its forward-looking claims should be separated from demonstrated capabilities and read with the limitations of an unreviewed preprint.

https://arxiv.org/pdf/2507.11552

18 Constitutional AI explained

Toloka AI / Industry explainer **GRADE C**

A practitioner-oriented explanation of constitutional principles, self-critique, revision, and AI feedback. It is useful for a process overview, but it should be paired with the original technical paper when describing algorithms, experiments, or comparative performance.

https://toloka.ai/blog/constitutional-ai-explained/

19 Do Moral Judgment and Reasoning Capability of LLMs Change with Language? A Study using the Multilingual Defining Issues Test

Proceedings of EACL 2024 / Peer-reviewed conference paper **GRADE A**

This study applies a multilingual Defining Issues Test to ChatGPT, GPT-4, and Llama2Chat-70B across English and five additional languages. It is the report's strongest source for language-dependent variation in model judgments and post-conventional scores; it also cautions that psychometric outputs do not establish human-like moral cognition.

https://aclanthology.org/2024.eacl-long.176.pdf

Kohlberg applied to AI and multilingual DIT studies

20 Microsoft researchers propose DIT morality test for LLMs

Reddit / r/MachineLearning / Discussion thread **GRADE D**

A community discussion pointing readers toward research on the Defining Issues Test for language models. It can reveal practitioner reactions or locate the underlying paper, but comments are not vetted evidence and should never substitute for the study itself.

https://www.reddit.com/r/MachineLearning/comments/16t52u7/r_microsoft_researchers_propose_dit_morality_test/

21 Language Model Moral Competence

MINT Lab / Research-group report **GRADE B**

A research-group synthesis of what it means to evaluate moral competence in language models. It is useful for distinguishing behavior, reasoning, and competence, though any empirical result cited from the page should be traced to the underlying paper or dataset.

https://mintresearch.org/reports/language-model-moral-competence/

22 Daily Papers: morality briefing

Hugging Face / Paper-discovery index **GRADE D**

A dynamic search or discovery page rather than a stable scholarly item. It may help find relevant preprints, but it does not support a specific claim and should be replaced in a formal bibliography by the individual papers selected from it.

https://huggingface.co/papers?q=morality%20briefing

Kohlberg applied to AI and multilingual DIT studies

23 Review: Do Moral Judgment and Reasoning Capability of LLMs Change with Language?

Moonlight / Automated / secondary paper review **GRADE C**

A secondary summary of the EACL multilingual DIT study. It may help readers quickly understand the design, but the peer-reviewed paper is the controlling source for methods, statistics, and limitations.

https://www.themoonlight.io/fr/review/do-moral-judgment-and-reasoning-capability-of-llms-change-with-language-a-study-using-the-multilingual-defining-issues-test

Moral-reasoning benchmarks

24 LLM Ethics Benchmark: A Three-Dimensional Assessment System for Evaluating Moral Reasoning in Large Language Models

arXiv / Preprint with public code/data **GRADE B**

This paper proposes three dimensions: foundational moral principles, reasoning robustness, and value consistency. It directly supports the report's benchmark table and conceptual framing, but its reported scores should be treated as benchmark-specific and verified against the paper's model versions, prompts, and evaluation code.

https://arxiv.org/html/2505.00853v1

25 Introducing the LLM Ethics Benchmark

University of Texas at Austin, Ethical AI / University research announcement **GRADE B**

A university announcement describing the purpose and release of the LLM Ethics Benchmark. It is appropriate for provenance and project context, while technical claims and numerical results should cite the associated paper and repository.

https://ethicalai.utexas.edu/2025/05/16/new-research-from-urban-information-lab-introducing-the-llm-ethics-benchmark/

26 ETHICS Benchmark for AI Ethics

Emergent Mind / Research-topic explainer **GRADE C**

A synthesized overview of ethics benchmarks and related literature. It can orient readers to tasks and terminology, but it is a secondary aggregator; claims about dataset size, categories, or model results should be checked against the original ETHICS paper and dataset card.

https://www.emergentmind.com/topics/ethics-benchmark

Moral-reasoning benchmarks

27 SafetyPrompts.com

SafetyPrompts.com / Prompt collection / resource site **GRADE D**

A broad resource for safety-related prompts rather than a clearly specified moral-reasoning benchmark. Its relevance to Kohlbergian assessment is indirect, and reproducible evaluation requires a fixed dataset, protocol, scoring rule, and versioned release.

https://safetyprompts.com/

28 ProgressGym: Alignment with a Millennium of Moral Progress

arXiv / Preprint **GRADE B**

A benchmark and training framework built around historical moral change. It broadens evaluation beyond static contemporary norms, but its notion of progress embeds normative assumptions that should be made explicit when comparing it with Kohlberg's developmental hierarchy.

https://arxiv.org/html/2406.20087v2

29 Structured Moral Reasoning in Language Models

ACL Anthology / Peer-reviewed conference paper **GRADE A**

A technical study of methods for eliciting or evaluating structured moral reasoning in language models. It is a strong source for moving beyond answer-only classification, although readers should still separate improved reasoning traces from evidence of stable moral agency.

https://aclanthology.org/2025.emnlp-main.1541.pdf

Moral-reasoning benchmarks

30 Datasets

Vulcan Lab / Research-lab dataset index **GRADE C**

An index of datasets produced or curated by a research group. It is useful for discovery and access; each dataset's own paper, license, documentation, and version should be cited for task construction and results.

https://vulcanlab.ai/research/datasets/

31 MTMA: Multi-Turn RL for Long-Horizon Moral Alignment

OpenReview / Conference submission / review record **GRADE B**

A paper on reinforcement learning for moral behavior across extended interactions. It is relevant because long-horizon consistency is harder than one-shot dilemma answering; interpret its status and findings in light of the OpenReview decision, reviews, and final version.

https://openreview.net/pdf?id=BhgpIz5okJ

32 LLM Ethics Benchmark: A Three-Dimensional Assessment System for Evaluating Moral Reasoning in Large Language Models

ResearchGate / Mirror of preprint **GRADE C**

A ResearchGate copy of the same benchmark represented by items 24 and 33. It adds no independent evidence; cite the arXiv record or a published version so readers can identify the canonical text and version.

https://www.researchgate.net/publication/391443969_LLM_Ethics_Benchmark_A_Three-Dimensional_Assessment_System_for_Evaluating_Moral_Reasoning_in_Large_Language_Models

Moral-reasoning benchmarks

33 LLM Ethics Benchmark, arXiv:2505.00853v1

arXiv / Preprint PDF **GRADE B**

The PDF form of the three-dimensional benchmark paper. It is a canonical, stable citation for the preprint, but this entry duplicates item 24; one canonical record is enough in a cleaned bibliography.

https://arxiv.org/pdf/2505.00853

34 Evaluating Moral Beliefs across LLMs through a Pluralistic Framework

arXiv / Preprint **GRADE B**

A pluralistic evaluation framework that avoids treating a single moral theory as the sole ground truth. It is important for qualifying Kohlberg-centered comparisons, since high performance under one schema may obscure disagreement across cultures, stakeholders, and ethical traditions.

https://arxiv.org/html/2411.03665v1

Alignment engineering and public constitutions

35 Direct Preference Optimization: A Deep Dive into RLHF Alternatives

Medium / Technical blog **GRADE C**

A detailed secondary tutorial on Direct Preference Optimization (DPO). It may clarify intuition and implementation, but mathematical claims should be checked against the original DPO paper and trusted code because Medium posts are not peer reviewed.

https://medium.com/@vivekmgpr/direct-preference-optimization-a-technical-deep-dive-into-the-post-rlhf-era-of-llm-alignment-25f357f0d9b3

36 RLHF and alternatives: DPO and CoH

Argilla / Industry technical blog **GRADE C**

A practical comparison of reinforcement learning from human feedback (RLHF), DPO, and related approaches. It is useful for implementation context, though it should not be the sole source for formal objectives, convergence claims, or benchmark comparisons.

https://argilla.io/blog/mantisnlp-rlhf-part-3/

37 Direct-Alignment Algorithms

RLHF Book by Nathan Lambert / Open technical textbook **GRADE B**

A technical chapter that situates DPO among direct-alignment methods and explains the preference-learning objective. It is a stronger pedagogical source than a general blog, while original papers remain preferable for attribution and experimental claims.

https://rlhfbook.com/c/08-direct-alignment

Alignment engineering and public constitutions

38 Constitutional AI: How Anthropic Trains Models Using Written Principles

TDWI / Industry explainer **GRADE C**

A recent industry explanation of Constitutional AI and written principles. It is suitable for nontechnical orientation, but first-party and peer-reviewed sources should ground the training pipeline, safety results, and model-behavior claims.

https://tdwi.org/blogs/ai-101/2026/05/constitutional-ai.aspx

39 Claude's New Constitution: AI Alignment, Ethics, and the Future of Model Governance

British & Irish Studies in Intelligence / Policy commentary **GRADE C**

A governance-oriented interpretation of a model constitution and its institutional implications. It can enrich policy discussion, but it is commentary; the constitution itself is the primary source for what principles the model developer states.

https://bisi.org.uk/reports/claudes-new-constitution-ai-alignment-ethics-and-the-future-of-model-governance

40 AI Alignment Takes us Toward a Global Ethic

Markkula Center for Applied Ethics, Santa Clara University / Ethics commentary **GRADE B**

An ethics-center essay linking alignment to shared global principles. Its value lies in normative framing and questions about pluralism, not empirical evidence that current models converge on a universal ethic.

https://www.scu.edu/ethics/leadership-ethics-blog/ai-alignment-takes-us-toward-a-global-ethic/

Alignment engineering and public constitutions

41 Collective Constitutional AI: Aligning a Language Model with Public Input

ACM FAccT 2024 / arXiv / Peer-reviewed conference paper **GRADE A**

This paper describes a process for eliciting public principles, forming a constitution, training a model, and evaluating resulting behavior. It provides concrete evidence for participatory alignment, while explicitly leaving open how to measure full adherence and how a chosen public represents wider populations.

https://arxiv.org/html/2406.07814v1

Moral rhetoric, corrigibility, and model constitutions

42 Reasoning or Rhetoric? An Empirical Analysis of Moral Reasoning in Language Models

arXiv / Preprint **GRADE B**

An empirical inquiry into whether moral-sounding model outputs reflect robust reasoning or rhetorical pattern reproduction. It is central to the report's moral-ventriloquism concern, but conclusions should be tied to the tested models, prompts, and operational definition of reasoning.

https://arxiv.org/abs/2603.21854

43 Claude Wants to Kill You: Why Anthropic's Constitutional AI Will Fail

Medium / Polemic / commentary **GRADE D**

A provocative critique of Constitutional AI. It may surface failure hypotheses worth testing, but the sensational framing and lack of peer review mean it should be treated as an argument, not evidence of actual model intent, capability, or observed harm.

https://medium.com/@gp2030/claude-wants-to-kill-you-21b72ffa2de7

44 Claude's Constitution

Anthropic / First-party policy document **GRADE A**

Anthropic's first-party statement of principles intended to shape Claude's behavior. It is authoritative for the developer's stated normative framework, but not independent evidence that the deployed model consistently follows those principles or that the principles resolve value conflicts.

https://www.anthropic.com/constitution

Moral rhetoric, corrigibility, and model constitutions

45 Artificial Intelligence 2025 listing

arXiv / Category index **GRADE D**

A broad arXiv listing page rather than a specific work. It cannot support any identifiable claim and should be removed from a cleaned bibliography unless the intended paper is located and cited directly.

https://www.arxiv.org/list/cs.AI/2025?skip=27100&show=2000